

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Kyle tried to use the table of integrals to evaluate the integral $\int (5-4x^2)^{3/2} dx$. His work is shown below. His final answer is not correct. What did he do wrong?

Kyle's Work:

We use entry #92 from the table of integrals in our textbook which states that

$$\int (a^2 - u^2)^{3/2} du = -\frac{u}{8}(2u^2 - 5a^2)\sqrt{a^2 - u^2} + \frac{3a^4}{8}\sin^{-1}\left(\frac{u}{a}\right) + C \quad \text{if } a > 0.$$

Clearly here $a = \sqrt{5}$ & $u = 2x$, so this means that

$$\int (5-4x^2)^{3/2} dx = -\frac{2x}{8}\left(2(2x)^2 - 5(\sqrt{5})^2\right)\sqrt{5-4x^2} + \frac{3(\sqrt{5})^4}{8}\sin^{-1}\left(\frac{2x}{\sqrt{5}}\right) + C$$

The issue here is that Kyle didn't fully convert the given integral into the one from the table of integrals.

To do this note that $a = \sqrt{5}$, $u = 2x$, & $du = 2dx \rightarrow \frac{1}{2}du = dx$, and therefore,

$$\begin{aligned} \int (5-4x^2)^{3/2} dx &= \int (a^2 - u^2)^{3/2} \frac{1}{2} du = \frac{1}{2} \int (a^2 - u^2)^{3/2} du \\ &= \frac{1}{2} \left[-\frac{2x}{8}\left(2(2x)^2 - 5(\sqrt{5})^2\right)\sqrt{5-4x^2} + \frac{3(\sqrt{5})^4}{8}\sin^{-1}\left(\frac{2x}{\sqrt{5}}\right) \right] + C \end{aligned}$$

2. Use an entry from the table of integrals to evaluate the integral $\int \frac{dx}{\sqrt{2x^3+3x^2}}$, for $x > 0$.

Note that $\int \frac{dx}{\sqrt{2x^3+3x^2}} = \int \frac{dx}{\sqrt{x^2(2x+3)}} = \int \frac{dx}{|x|\sqrt{2x+3}} = \int \frac{dx}{x\sqrt{2x+3}}.$

We then use entry #108 from the table of integrals in our textbook which states that

$$\int \frac{du}{u\sqrt{a+bu}} = \frac{1}{\sqrt{a}} \ln \left| \frac{\sqrt{a+bu} - \sqrt{a}}{\sqrt{a+bu} + \sqrt{a}} \right| + C \quad \text{if } a > 0.$$

We have that $a = 3$, $b = 2$, $u = x$, & $du = dx$, therefore,

$$\int \frac{dx}{x\sqrt{2x+3}} = \int \frac{du}{u\sqrt{a+bu}} = \frac{1}{\sqrt{a}} \ln \left| \frac{\sqrt{a+bu} - \sqrt{a}}{\sqrt{a+bu} + \sqrt{a}} \right| + C = \frac{1}{\sqrt{3}} \ln \left| \frac{\sqrt{3+2x} - \sqrt{3}}{\sqrt{3+2x} + \sqrt{3}} \right| + C$$

3. Evaluate the integral $\int_1^e \frac{1}{x\sqrt{4+\ln^2(x)}} dx$ using an entry from the table of integrals.

We use entry #72 from the table of integrals in our textbook which states that

$$\int \frac{du}{\sqrt{a^2+u^2}} = \ln(u + \sqrt{a^2+u^2}) + C.$$

Clearly here $a=2$ & $u=\ln(x)$, so this means that $du = \frac{1}{x} dx$ and the bounds become

$u(1)=0$ & $u(e)=1$ Therefore we have,

$$\begin{aligned} \int_1^e \frac{1}{x\sqrt{4+\ln^2(x)}} dx &= \int_0^1 \frac{du}{\sqrt{a^2+u^2}} = \ln(u + \sqrt{a^2+u^2}) \Big|_0^1 \\ &= \ln(1 + \sqrt{a^2+1}) - \ln(0 + \sqrt{a^2+0}) \\ &= \ln(1 + \sqrt{5}) - \ln(\sqrt{4}) \\ &= \ln\left(\frac{1+\sqrt{5}}{2}\right) \end{aligned}$$

4. Entry #93b in the table of integrals from our textbook states that $\int \frac{du}{a^2-u^2} = \frac{1}{2a} \ln \left| \frac{u+a}{u-a} \right| + C$. Prove that this is correct by fully integrating and simplifying $\int \frac{du}{a^2-u^2}$.

$$\int \frac{du}{a^2-u^2} = \int \frac{du}{(a-u)(a+u)}.$$

By partial fractions we see that

$$\frac{1}{(a-u)(a+u)} = \frac{A}{a-u} + \frac{B}{a+u} = \frac{A(a+u)+B(a-u)}{(a-u)(a+u)}$$

We need

$$\begin{cases} A(a+u)+B(a-u)=1 \\ (A-B)u+a(A+B)=1 \end{cases} \text{ which implies that } \begin{cases} A-B=0 \\ A+B=\frac{1}{a} \end{cases} \rightarrow 2A=\frac{1}{a} \rightarrow A=\frac{1}{2a} \rightarrow B=\frac{1}{2a}$$

Therefore, we have

$$\begin{aligned} \int \frac{du}{a^2-u^2} &= \int \frac{du}{(a-u)(a+u)} = \int \left[\frac{1}{2a(a-u)} - \frac{1}{2a(a+u)} \right] du \\ &= \frac{1}{2a} \ln|a-u| - \frac{1}{2a} \ln|a+u| + C = \frac{1}{2a} [\ln|a-u| - \ln|a+u|] + C \\ &= \frac{1}{2a} \ln \left| \frac{a-u}{a+u} \right| + C = \frac{1}{2a} \ln \left| -\frac{u-a}{u+a} \right| + C \\ &= \frac{1}{2a} \ln \left| \frac{u-a}{u+a} \right| + C \end{aligned}$$

5. Suppose that the trig substitution $x = \sec(\theta)$ is being used on an integral where $x < 0$.
- a) How does the fact that $x < 0$ allow you to change that angle restriction that would need to put on θ ? Determine the smallest angle restriction that can be used for θ .

Our standard angle restriction would be $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$.

If we know that $x < 0$, then it must be the case that $\sec(\theta) < 0$ but based on the angle restriction that we have it would have to be the case that $\pi \leq \theta < \frac{3\pi}{2}$.

Therefore, $\pi \leq \theta < \frac{3\pi}{2}$ would be the angle restriction that we would have to use.

- b) Based on the angle restriction that you established in part (a) determine whether or not the statement $\theta = \sec^{-1}(x)$ is true or false.

We know that $\sec^{-1}(x)$ can only return angles that exist within $[0, \pi]$ and therefore it can never return the angle θ since $\pi \leq \theta < \frac{3\pi}{2}$.

- c) If $\pi \leq \theta < \frac{3\pi}{2}$ and $x < 0$, then the statement $\theta = C - \sec^{-1}(x)$ is always true (where C is some specific constant). Determine the value of C which makes this statement true.

Note that if $x < 0$ then $\sec^{-1}(x)$ will return an angle in the 2nd quadrant. We would like to modify this formula so that it returns an angle in the 3rd quadrant.

The angle in the 3rd quadrant that returns the same secant value as the angle in the 2nd quadrant has the same distance angle measure but is negative.

Therefore, we can say that $\theta = 2\pi - \sec^{-1}(x)$.

6. Determine the integral $\int \frac{x^2 - 1}{x\sqrt{x^2 - 4}} dx$ assuming that $x < 0$.

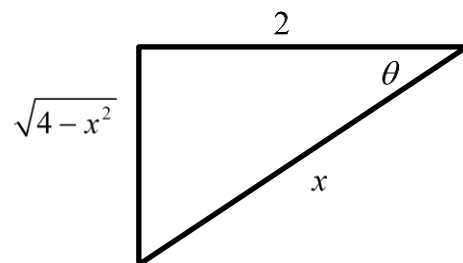
Let $x = 2\sec(\theta)$, where $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$, then $dx = 2\sec(\theta)\tan(\theta)d\theta$. Since $x < 0$ we can further restrict the angle restriction to $\pi \leq \theta < \frac{3\pi}{2}$.

$$\begin{aligned} \int \frac{x^2 - 1}{x\sqrt{x^2 - 4}} dx &= \int \frac{\sec^2(\theta) - 1}{4\sec(\theta)\sqrt{4\sec^2(\theta) - 4}} \sec(\theta)\tan(\theta)d\theta = \int \frac{\sec^2(\theta) - 1}{8\sqrt{\tan^2(\theta)}} \tan(\theta)d\theta \\ &= \int \frac{1}{8} [\sec^2(\theta) - 1] d\theta \\ &= \frac{1}{8} \tan(\theta) - \theta + C \end{aligned}$$

We must now write our solution in terms of x . Since

$$x = 2\sec(\theta) \rightarrow \sec(\theta) = \frac{x}{2} \text{ and due to the angle restriction, we}$$

can say that $\sec(\theta) = \frac{x}{2} \rightarrow \theta = 2\pi - \sec^{-1}\left(\frac{x}{2}\right)$. We can construct the following reference triangle for θ .



From the reference triangle we see that $\tan(\theta) = \frac{\sqrt{4 - x^2}}{x}$ and therefore

$$\begin{aligned} \frac{1}{8} \tan(\theta) - \theta + C &= \frac{1}{8} \left(\frac{\sqrt{4 - x^2}}{x} \right) - \left[2\pi - \sec^{-1}\left(\frac{x}{2}\right) \right] + C \\ &= \frac{\sqrt{4 - x^2}}{8x} - 2\pi + \sec^{-1}\left(\frac{x}{2}\right) + C \\ &= \frac{\sqrt{4 - x^2}}{8x} + \sec^{-1}\left(\frac{x}{2}\right) + C_1 \end{aligned}$$